

five technologies and strategic initiatives in business intelligence. Big Data currently ranks 20th across 33 key technologies. Big Data analytics has greater strategic importance than IoT, natural language processing, cognitive business intelligence (BI) and location intelligence.

- Among the 53 per cent of the enterprises worldwide that are currently using data science for decision making, telecom and financial sector companies are major adopters of Big Data analytics.
- Data warehouse optimisation is regarded as a very important aspect of Big Data analytics, followed by customer analysis and predictive maintenance.
- Spark, MapReduce and Yarn are the most popular data science analytics software frameworks today.
- The most popular Big Data access methods include: Spark SQL, Hive, HDFS and Amazon S3. A 73 per cent share of this segment is held by Spark SQL, followed by 26 per cent by Hive and HDFS, and the rest by Amazon S3.
- Machine learning is gaining more industrial support with the Spark Machine Learning Library (MLlib) and the adoption rate is expected to rise by 60 per cent in the next one year.

Considering the adoption rate of data science in enterprises, lots of commercial and open source tools are now available. The focus of this article is to cover the various open source tools for data science, especially those based on machine learning and deep learning.

1. Apache Mahout

The Apache Mahout project was started by a group of people involved in Apache Lucene, who conducted a lot of research in machine learning and wanted to develop robust, well documented, scalable implementations of common machine learning algorithms for clustering and categorisation. The primary objective behind the

development of Mahout was to:

- Build and support a community of users to contribute source code to Apache Mahout.
- Focus on real-world, practical use cases as compared to bleeding-edge research or unproven techniques.
- Strongly support the software with documentation and examples.

Apache Mahout is a scalable machine learning library. It is available as open source software under the Apache Software Foundation. It supports algorithms for clustering, classification and collaborative filtering on distributed platforms. Mahout also provides Java libraries for common maths operations (focused on linear algebra and statistics) and primitive Java collections. It has a few algorithms that are implemented as MapReduce. These algorithms can run on Hadoop to exploit the parallelism in a distributed cluster. Once Big Data is stored in the Hadoop Distributed File System (HDFS), Mahout provides data science tools to automatically find meaningful patterns in big data sets.

Features

- Allows collaborative filtering, i.e., it mines user behaviour and makes probable product recommendations.
- Allows clustering, which organises items in a particular class, and naturally occurring groups in such a way that all the items belonging to the same group share similar characteristics.
- Allows automatic classification by learning from existing categories and accordingly assigns unclassified items to the best category.
- Another delightful feature is frequent item set mining that analyses items in a group, i.e., those similar to items in a shopping cart and then identifies which items typically appear together.
- Incorporates a number of pre-designed algorithms for Scala + Apache Spark, H2O, Apache Flink, etc.
- Contains Samsara, which is a vector

maths experimentation environment with R-like syntax, which works at scale.

- Includes several MapReduce enabled clustering implementations like k-means, fuzzy k-means, Canopy, Dirichlet and MeanShift.
- Comes with distributed fitness function capabilities for evolutionary programming.
- Includes matrix and vector libraries.

Official website: <https://mahout.apache.org/>

Latest version: 0.13.0

2. Apache SystemML

Apache SystemML was designed by researchers at the IBM Almaden Research Center in 2010. Previously, when data scientists wrote machine learning algorithms using R and Python for small data and for Big Data, they made use of SCALA. This process was long, full of iterations and sometimes error prone. In order to simplify the process, SystemML was proposed. The objective of SystemML was to automatically scale an algorithm written in R or Python to handle Big Data without any errors by using the multi-iterative translation approach.

Apache SystemML is a new, flexible machine learning system that automatically scales to Spark and Hadoop clusters. It provides a high-level language to quickly implement and run machine learning algorithms on Spark. It advances machine learning in two important ways – the Apache SystemML language and Declarative Machine Learning (DML) – and includes linear algebra primitives, statistical functions and ML constructs. It provides automatic optimisation as per data and cluster to ensure efficiency and scalability.

Features

- *Customisation of algorithms:* All machine learning algorithms in SystemML are in high-level, declarative machine learning language (DML). This language